Sales Analytics Dashboard

1. Project Overview

This project is an end-to-end sales analytics project designed to transform raw transactional sales data into actionable business insights.

The project analyses sales transactions across products, regions, sales representatives, customers, sales channels, payment methods and discount levels.

The analysis was conducted using **PostgreSQL, SQL, DAX and Power BI**, following a workflow from data preparation and validation through to interactive dashboard development and business recommendations.

The primary objective was to understand sales performance, identify high-performing areas, evaluate customer behaviour and investigate the relationship between discounts and revenue.

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2. Business Problem

A business needs to understand how its sales are performing across different products, regions, sales representatives and customer segments.

Raw transaction data alone does not provide an efficient way for decision-makers to identify trends or areas requiring attention.

This project therefore aims to answer questions such as:

- How much revenue has been recorded?
- How many transactions have occurred?
- How many units have been sold?
- Which products and categories generate the most revenue?
- Which regions perform best?
- Which sales representatives generate the most revenue?
- How do online and retail sales compare?
- Which payment methods generate the most revenue?
- How do new and returning customers compare?
- Which customer types generate the most revenue?
- How do discount levels relate to revenue?
- Which areas of the business should management investigate further?

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3. Dataset

The dataset contains transactional sales information covering products, customers, sales representatives and sales activity.

The main fields used in the analysis include:

- Product ID
- Sale Date
- Sales Representative
- Region
- Product Category
- Customer Type
- Sales Channel
- Payment Method
- Quantity Sold
- Sales Amount
- Unit Cost
- Unit Price
- Discount

The original dataset contains **1,000 transaction records**.

For the dashboard's baseline reporting, the project uses the recorded `sales_amount` field as the revenue measure.

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4. Tools and Technologies

Tool	Purpose
PostgreSQL	Data storage, validation and analysis
SQL	Data exploration, aggregation and validation
DAX	Power BI measures and analytical calculations
Power BI	Dashboard development and data visualisation
Excel/CSV	Initial data handling

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5. Project Workflow

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Raw Sales Dataset
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Data Preparation
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        v
PostgreSQL
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        v
SQL Validation and Analysis
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        v
Reporting View
        |
        v
Power BI Data Model
        |
        v
DAX Measures
        |
        v
Interactive Dashboard
        |
        v
Business Insights
        |
        v
Recommendations
```

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6. Data Preparation and Validation

The raw sales data was imported into PostgreSQL for structured analysis.

SQL was used to:

- Inspect the dataset
- Validate data types
- Check transaction records
- Analyse missing or inconsistent values
- Aggregate sales performance
- Segment customers
- Analyse products and regions
- Investigate discounts
- Prepare the data for Power BI

A reporting view was subsequently used to provide Power BI with a structured analytical dataset.

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7. Key Performance Indicators

The dashboard uses the following primary KPIs:

Total Transactions

1,000

This represents the number of source transaction records in the dataset.

Recorded Revenue

5,019,265.23

This represents the total recorded sales revenue based on the `sales_amount` field.

Units Sold

25,355

This represents the total quantity of products sold across the dataset.

Average Transaction Value

5,019.27

This represents the average recorded sales amount per transaction.

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8. Important Revenue Definition

The dashboard uses **recorded sales revenue** rather than calculated profit as the primary financial KPI.

This decision was made because inconsistencies were identified between the cost and revenue-related fields in the dataset.

As a result:

- Revenue analysis is included.
- Transaction analysis is included.
- Product and category sales analysis is included.
- Profitability analysis is excluded from the final dashboard.

This prevents potentially unreliable cost calculations from being presented as accurate business profitability.

9. Power BI Dashboard

The Power BI dashboard consists of three pages.

Page 1 — Executive Overview

The first page provides a high-level summary of overall sales performance.

KPIs

- Recorded Revenue
- Total Transactions
- Units Sold
- Average Transaction Value

Key Visuals

- Revenue Trend Over Time
- Revenue by Product Category
- Revenue by Region

Purpose

The Executive Overview allows a stakeholder to quickly understand the overall performance of the business before exploring more detailed analysis.

Page 2 — Sales Performance

The second page focuses on sales performance across organisational and transactional dimensions.

Key Visuals

- Revenue by Region and Product
- Revenue by Sales Representative
- Revenue by Sales Channel
- Revenue by Payment Method

Purpose

This page allows management to compare sales performance across regions, products, representatives and transaction channels.

It can help identify high-performing areas and areas that may require further investigation.

Page 3 — Customer and Discount Insights

The third page focuses on customer behaviour and the relationship between discounts and sales.

Key Visuals

- New vs Returning Customer Revenue
- Revenue by Customer
- Customer Type by Product Category
- Revenue by Discount Band

Purpose

This page examines customer purchasing behaviour and evaluates how different discount levels are associated with recorded revenue.

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10. Key Findings

10.1 Product Category Performance

Clothing was the leading product category by recorded revenue.

This suggests that Clothing represents an important contributor to overall sales performance and may warrant further investigation into product-level performance, demand and customer preferences.

10.2 Regional Performance

The **North region** generated the highest share of recorded revenue at approximately **27.29%**.

Regional performance therefore varies, creating an opportunity to investigate why some regions generate stronger sales than others.

Potential areas for further investigation include:

- Product mix
- Customer composition
- Sales representatives
- Regional demand
- Sales channels

- Discounting strategies

10.3 Customer Type

The customer base is relatively balanced between new and returning customers.

Returning customers generated slightly higher recorded revenue than new customers.

This suggests that customer retention contributes meaningfully to sales performance.

10.4 Product Preferences by Customer Type

Different customer types show different category preferences.

Among new customers, **Clothing** was the leading category.

Among returning customers, **Furniture** was the leading category.

This difference could be useful when developing customer-specific marketing and sales strategies.

10.5 Discount Analysis

The **20–30% discount band** generated the highest recorded revenue.

This indicates that transactions within this discount range contributed the largest amount of revenue in the dataset.

However, higher revenue does not automatically mean that the discount strategy is more profitable.

Because the project's profitability analysis was excluded due to inconsistencies in cost and revenue fields, the results should be interpreted as a **revenue association rather than proof that 20–30% discounts are optimal for profitability**.

10.6 Sales Channel

Retail generated slightly higher recorded revenue than Online.

However, the Online channel produced a slightly higher average transaction value.

This suggests that the channels may have different purchasing patterns and should be evaluated using multiple performance metrics rather than revenue alone.

10.7 Payment Method

Credit Card generated the highest recorded revenue among the payment methods analysed.

This provides an indication of the payment preferences associated with higher sales volumes in the dataset.

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11. Business Recommendations

11.1 Investigate Regional Performance

The North region generated the largest share of recorded revenue.

Management could investigate what contributes to this performance and determine whether successful practices can be replicated in lower-performing regions.

11.2 Analyse Clothing Performance

Since Clothing is the leading category, the business could investigate:

- Best-selling products
- Customer segments purchasing Clothing
- Regional demand
- Seasonal patterns
- Discount levels
- Sales representatives associated with the category

11.3 Strengthen Customer Retention

Returning customers generate slightly higher revenue than new customers.

This suggests that customer retention may be an important contributor to revenue.

The business could investigate loyalty programmes, repeat-purchase campaigns and personalised offers.

11.4 Review Discount Strategy

The 20–30% discount band generated the highest recorded revenue.

However, the business should not automatically increase discounts.

Further analysis should determine whether the additional revenue generated by discounts is sufficient to compensate for the reduction in selling price.

A future version of the project could perform a reliable profitability analysis once the underlying cost data has been validated.

11.5 Compare Sales Channels

Retail generates slightly higher revenue, while Online has a slightly higher average transaction value.

Management should therefore evaluate the two channels using:

- Revenue
- Transaction volume
- Average transaction value
- Units sold
- Customer type
- Product category

This could identify different customer behaviours across channels.

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12. Limitations

Profitability Analysis

Profit analysis was excluded because inconsistencies were identified between the cost and revenue fields.

Therefore, this project focuses on **recorded sales revenue rather than profit**.

Revenue Does Not Equal Profit

A category, region or discount band generating higher revenue is not necessarily more profitable.

Association Does Not Imply Causation

The analysis identifies patterns and relationships in the dataset but does not establish that a particular factor directly causes higher or lower sales.

Dataset Scope

The conclusions are based on the available transaction data and may not represent the complete operational performance of a real business.

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13. Future Improvements

Future versions of this project could include:

- Validated profitability analysis
- Gross profit and margin calculations
- Product-level profitability
- Customer lifetime value
- Customer segmentation
- Sales forecasting
- Time-series forecasting
- Sales representative performance targets
- Regional performance benchmarking
- Discount elasticity analysis
- Customer churn analysis
- Automated Power BI refreshes
- Predictive sales modelling using Python

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14. Skills Demonstrated

Data Analytics

- Exploratory Data Analysis
- Sales Performance Analysis
- Customer Analysis
- Revenue Analysis
- Segmentation
- KPI Development
- Business Insight Generation

Technical Skills

- SQL
- PostgreSQL
- DAX
- Power BI
- Data Cleaning
- Data Validation

- Data Transformation
- Data Visualisation
- Dashboard Development

Business Analysis

- Business Problem Definition
- KPI Selection
- Stakeholder-Oriented Reporting
- Identifying Performance Drivers
- Translating Data into Business Insights
- Developing Data-Driven Recommendations

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15. Conclusion

This project demonstrates an end-to-end approach to sales analytics, from raw transactional data through SQL-based analysis to an interactive Power BI dashboard.

The analysis identified important differences across product categories, regions, customer types, sales channels, payment methods and discount levels.

The findings indicate that **Clothing**, the **North region**, **returning customers**, and the **20–30% discount band** are important areas of the recorded revenue performance.

The project also demonstrates the importance of validating financial data before calculating profitability. Rather than presenting potentially unreliable profit figures, the dashboard focuses on the validated recorded sales revenue available in the dataset.

Overall, the project demonstrates how SQL and Power BI can be used to transform transactional data into clear business insights and support data-driven decision-making.